

Unit 9. Modelling Trends in Time

1 Trends in Time

FREES §7.1–7.2.

The data are now a single series $y_1, \dots, y_T$ indexed by time. The index is not a label: observations near each other in time are typically dependent, and the classical assumption of Unit 2 that the errors are uncorrelated is the first thing to check rather than the last.

1.1 Decomposition

Definition 9.1 (Components). A series is decomposed into a **trend** μ_t , a **seasonal** component γ_t of known period s , and an **irregular** component ε_t , either additively or multiplicatively:

$$y_t = \mu_t + \gamma_t + \varepsilon_t \quad \text{or} \quad y_t = \mu_t \gamma_t \varepsilon_t. \quad (1)$$

Periodicity of the seasonal component means $\gamma_{t+s} = \gamma_t$ for every t .

Proposition 9.2 (Logarithms turn one into the other). If $y_t = \mu_t \gamma_t \varepsilon_t$ with all three factors positive, then $\ln y_t = \ln \mu_t + \ln \gamma_t + \ln \varepsilon_t$ is additive, and $\ln \gamma_t$ has the same period s . A seasonal

swing that is a constant **proportion** of the level is therefore a constant **amount** on the log scale.

Proof. Take logarithms; periodicity is preserved because $\ln \gamma_{t+s} = \ln \gamma_t$. For the last statement, if $\gamma_t = 1 + \delta$ in one season then the seasonal effect on y_t is $\delta\mu_t$, proportional to the level, while on the log scale it is $\ln(1 + \delta)$, free of t . $\square$

1.2 Polynomial trends

Definition 9.3 (Linear trend in time). The **linear trend in time** model is

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t, \quad t = 1, \dots, T, \quad (2)$$

with $\{\varepsilon_t\}$ uncorrelated, mean zero, variance σ^2 . Higher-order polynomials add $t^2, t^3, \dots$ as further regressors.

Theorem 9.4 (Exact least squares for a linear trend). For the design $\mathbf{X} = [\mathbf{1} \quad \mathbf{t}]$ with $\mathbf{t} =$

$$(1, 2, \dots, T)',$$

$$\text{var}(\hat{\beta}_1) = \frac{12 \sigma^2}{T(T^2 - 1)}, \quad \text{var}(\hat{\beta}_0) = \frac{2(2T + 1) \sigma^2}{T(T - 1)}, \quad \text{cov}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-6 \sigma^2}{T(T - 1)}. \quad (3)$$

Proof. With $\sum_t t = T(T + 1)/2$ and $\sum_t t^2 = T(T + 1)(2T + 1)/6$,

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} T & \frac{T(T+1)}{2} \\ \frac{T(T+1)}{2} & \frac{T(T+1)(2T+1)}{6} \end{pmatrix}, \quad \det(\mathbf{X}'\mathbf{X}) = \frac{T^2(T + 1)(2T + 1)}{6} - \frac{T^2(T + 1)^2}{4}.$$

Factoring $T^2(T + 1)/12$ out of the determinant leaves $2(2T + 1) - 3(T + 1) = T - 1$, so $\det(\mathbf{X}'\mathbf{X}) = T^2(T + 1)(T - 1)/12 = T^2(T^2 - 1)/12$. By Unit 2, $\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, and inverting a 2×2 matrix,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{\det} \begin{pmatrix} \frac{T(T+1)(2T+1)}{6} & -\frac{T(T+1)}{2} \\ -\frac{T(T+1)}{2} & T \end{pmatrix}.$$

The lower right entry is $T \cdot 12 / \{T^2(T^2 - 1)\} = 12 / \{T(T^2 - 1)\}$; the upper left is $\{T(T + 1)(2T +$

$1)/6\} \cdot 12/\{T^2(T+1)(T-1)\} = 2(2T+1)/\{T(T-1)\}$; and the off-diagonal is $-\{T(T+1)/2\} \cdot 12/\{T^2(T+1)(T-1)\} = -6/\{T(T-1)\}$. $\square$

Remark. $\text{var}(\hat{\beta}_1)$ falls like T^{-3} , not T^{-1} : the design points spread out as the series lengthens, so a slope in time is estimated far more precisely than a slope on a covariate whose spread is fixed. That precision is real only if the errors are genuinely uncorrelated, which for a time series is the assumption most likely to fail (§2).

Remark. A polynomial fits the observed range and then leaves it. The leading term dominates outside the data, so a high-degree fit that improves R^2 by a little can produce forecasts of any magnitude and sign (§5.1). Degree is a modelling choice made against out-of-sample error, not against R^2 .

1.3 Seasonality

Definition 9.5 (Seasonal dummies). For period s let $d_{tj} = \mathbf{1}\{t \equiv j \pmod{s}\}$, $j = 1, \dots, s$. The **seasonal dummy** model is $\gamma_t = \sum_{j=1}^s \alpha_j d_{tj}$.

Proposition 9.6 (Identifiability). The s dummy columns sum to $\mathbf{1}$, so a model containing all s of them and an intercept is rank deficient. Dropping either the intercept or one dummy restores full rank and leaves the column space, hence the fitted values, unchanged.

Proof. Every t lies in exactly one residue class, so $\sum_j d_{tj} = 1$ for all t and the $s + 1$ columns are dependent. The span of $\{\mathbf{1}, d_{\cdot 1}, \dots, d_{\cdot s}\}$ has dimension s , and each of the two reduced sets has s members that are linearly independent, so each spans it. Fitted values are the projection onto the column space (Unit 2), which is the same. $\square$

Proposition 9.7 (What the dummies fit). With the s dummies as the only regressors, $\hat{\alpha}_j$ is the mean of the observations in season j .

Proof. The columns are orthogonal, $d'_{\cdot j}d_{\cdot l} = 0$ for $j \neq l$, so the normal equations separate: $\hat{\alpha}_j = d'_{\cdot j}\mathbf{Y}/d'_{\cdot j}d_{\cdot j}$, which is the sum of the observations in season j divided by their number. $\square$

Theorem 9.8 (Harmonics span the same space). Fix $s \geq 2$ and let $\omega_j = 2\pi j/s$. The vectors

$$\mathbf{1}, \quad \{\cos(\omega_j t)\}_{j=1}^{\lceil s/2 \rceil - 1}, \quad \{\sin(\omega_j t)\}_{j=1}^{\lceil s/2 \rceil - 1}, \quad \text{and } \{\cos(\pi t)\} \text{ if } s \text{ is even,}$$

regarded as functions of t on one period, form a basis of the same s -dimensional space as the seasonal dummies.

Proof. Both families consist of functions of $t \bmod s$, so both lie in the space $\mathbb{R}^s$ of all such functions, of dimension s . Counting the harmonics: 1 constant, $2(\lceil s/2 \rceil - 1)$ sines and cosines, and one more when s is even, giving $1 + 2(s/2 - 1) + 1 = s$ for even s and $1 + 2\{(s+1)/2 - 1\} = s$ for odd s . It therefore suffices to show the harmonics are linearly independent.

Write $\zeta = e^{2\pi i/s}$. For integers a not divisible by s , $\sum_{t=0}^{s-1} \zeta^{at} = (\zeta^{as} - 1)/(\zeta^a - 1) = 0$, a geometric sum, while for $a \equiv 0$ the sum is s . Taking real and imaginary parts of $\zeta^{(j-k)t}$ and $\zeta^{(j+k)t}$ gives, for

$$1 \leq j, k < s/2,$$

$$\sum_{t=0}^{s-1} \cos(\omega_j t) \cos(\omega_k t) = \frac{s}{2} \delta_{jk}, \quad \sum_{t=0}^{s-1} \sin(\omega_j t) \sin(\omega_k t) = \frac{s}{2} \delta_{jk}, \quad \sum_{t=0}^{s-1} \cos(\omega_j t) \sin(\omega_k t) = 0,$$

and each of these is orthogonal to $\mathbf{1}$ because $\sum_t \cos(\omega_j t) = \sum_t \sin(\omega_j t) = 0$ for $j \neq 0$. When s is even, $\cos(\pi t) = (-1)^t$ has squared norm s and is orthogonal to the rest by the same computation. A set of nonzero mutually orthogonal vectors is linearly independent. $\square$

Remark. The two parametrisations fit identically. Harmonics are preferred when s is large and the seasonal shape is smooth, since dropping the high-frequency pairs costs few parameters and imposes smoothness; dummies are preferred when the seasonal pattern is irregular, since they impose nothing.

1.4 Forecasting from a trend model

Proposition 9.9 (Point forecast and interval). Under (2) with normal errors, the forecast of y_{T+l} is $\hat{y}_{T+l} = \hat{\beta}_0 + \hat{\beta}_1(T+l)$ and, writing $\mathbf{x}_{T+l} = (1, T+l)'$, the prediction interval of Unit 3 is

$$\hat{y}_{T+l} \pm t_{T-2, 1-\alpha/2} s \sqrt{1 + \mathbf{x}_{T+l}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_{T+l}}. \quad (4)$$

Its half-width grows with l , but only through the estimation term, and tends to ts times $\sqrt{1 + O(l^2/T^3)}$.

Proof. The forecast and the interval are the prediction results of Unit 3 applied at the design point $\mathbf{x}_{T+l}$. For the width, Theorem 9.4 gives

$$\mathbf{x}_{T+l}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_{T+l} = \frac{2(2T+1)}{T(T-1)} - \frac{12(T+l)}{T(T-1)} + \frac{12(T+l)^2}{T(T^2-1)},$$

each term of order T^{-1} or l^2T^{-3} . □

Remark. The interval (4) is dominated by the irreducible term 1 under the square root: for a trend model the uncertainty about a forecast ten steps ahead is barely larger than one step ahead. That is a strong claim, and it is false for the model of the next section, where the interval widens like $\sqrt{l}$. Which of the two is right is a question about the error process, not about the trend.

2 Stationarity and Random Walks

FREES §7.3–7.5.

2.1 Stationarity and white noise

Definition 9.10 (Stationarity). A series $\{y_t\}$ with finite second moments is **weakly stationary** if

$$\mathbb{E}[y_t] = \mu \text{ free of } t, \quad \text{cov}(y_t, y_{t+k}) = \gamma(k) \text{ free of } t, \quad (5)$$

and γ is its **autocovariance function**, $\rho(k) = \gamma(k)/\gamma(0)$ its **autocorrelation function**. It is **strictly stationary** if the joint law of $(y_{t_1}, \dots, y_{t_m})$ equals that of $(y_{t_1+h}, \dots, y_{t_m+h})$ for every choice of indices and every h .

Proposition 9.11. For a weakly stationary series, $\text{var}(y_t) = \gamma(0)$ is constant, $\gamma(-k) = \gamma(k)$, and $|\gamma(k)| \leq \gamma(0)$. A strictly stationary series with finite second moments is weakly stationary; the converse is false.

Proof. Taking $k = 0$ in (5) gives constant variance. Symmetry is $\text{cov}(y_t, y_{t-k}) = \text{cov}(y_{t-k}, y_t) = \gamma(k)$ after shifting the index by k . The bound is Cauchy–Schwarz (prerequisites) applied to $y_t - \mu$ and $y_{t+k} - \mu$, both of variance $\gamma(0)$. If the joint laws are shift invariant then so are the moments they determine. For the converse, weak stationarity constrains only the first two moments and says nothing about, for instance, the skewness of y_t . $\square$

Definition 9.12 (White noise). $\{\varepsilon_t\}$ is **white noise** if it is independent and identically distributed with mean μ_ε and variance $\sigma^2 < \infty$. Its autocovariance is $\gamma(0) = \sigma^2$ and $\gamma(k) = 0$ for $k \neq 0$.

Proposition 9.13 (Forecasting white noise). For white noise the best mean squared error forecast of y_{T+l} given $y_1, \dots, y_T$ is μ_ε for every $l \geq 1$, estimated by $\bar{y}$, and the forecast error variance $\sigma^2(1 + 1/T)$ does not depend on l .

Proof. y_{T+l} is independent of the past, so $\mathbf{E}[y_{T+l} \mid y_1, \dots, y_T] = \mu_\varepsilon$, which minimises mean squared error by the conditional expectation result of Unit 1. Replacing μ_ε by $\bar{y}$, the error $y_{T+l} - \bar{y}$ has variance $\sigma^2 + \sigma^2/T$, the two terms being independent. $\square$

Remark. White noise is the target rather than a model of anything: a filter is a procedure that reduces a series to white noise, and modelling is finished when nothing is left in the residuals that the ACF can see.

2.2 The random walk

Definition 9.14 (Random walk). Let $\{c_t\}$ be white noise with mean μ_c and variance σ_c^2 , and let y_0 be fixed. The **random walk with drift** μ_c is

$$y_t = y_{t-1} + c_t = y_0 + \sum_{j=1}^t c_j, \quad t \geq 1. \quad (6)$$

Theorem 9.15 (Moments). For the random walk (6) and $1 \leq r \leq t$,

$$\mathbb{E}[y_t] = y_0 + t\mu_c, \quad \text{cov}(y_r, y_t) = r\sigma_c^2, \quad \text{corr}(y_r, y_t) = \sqrt{r/t}. \quad (7)$$

In particular $\text{var}(y_t) = t\sigma_c^2$, so a random walk is not weakly stationary whenever $\sigma_c^2 > 0$; and for fixed lag k , $\text{corr}(y_{t-k}, y_t) = \sqrt{1 - k/t} \rightarrow 1$ as $t \rightarrow \infty$.

Proof. The mean is immediate from (6). For the covariance, with $r \leq t$,

$$\text{cov}(y_r, y_t) = \text{cov}\left(\sum_{j \leq r} c_j, \sum_{l \leq t} c_l\right) = \sum_{j \leq r} \sum_{l \leq t} \text{cov}(c_j, c_l) = \sum_{j \leq r} \sigma_c^2 = r\sigma_c^2,$$

only the diagonal terms surviving because the c_j are independent. Taking $r = t$ gives the variance, which depends on t , so (5) fails. The correlation is $r\sigma_c^2 / \sqrt{r\sigma_c^2 \cdot t\sigma_c^2} = \sqrt{r/t}$, and putting $r = t - k$ gives the last claim. $\square$

Remark. The last statement is the visual signature of a random walk: its sample autocorrelation function starts near 1 and decays hardly at all over the first many lags, because the population

correlations $\sqrt{1 - k/t}$ are all close to 1 when t is large.

2.3 Inference under a random walk

Theorem 9.16 (The sample mean is not what it looks like). For the driftless random walk, $\mu_c = 0$,

$$\text{var}(\bar{y}) = \frac{\sigma_c^2}{T^2} \sum_{m=1}^T m^2 = \sigma_c^2 \frac{(T+1)(2T+1)}{6T}, \quad (8)$$

which **grows** like $T\sigma_c^2/3$. The usual estimate s_y^2/T , computed as though the observations were independent, does not.

Proof. Write $y_t = y_0 + \sum_{j \leq t} c_j$. Then $\sum_{t=1}^T y_t = Ty_0 + \sum_{j=1}^T (T-j+1)c_j$, since c_j appears in y_t for every $t \geq j$, that is in $T-j+1$ of the terms. The c_j are independent, so

$$\text{var}\left(\sum_{t=1}^T y_t\right) = \sigma_c^2 \sum_{j=1}^T (T-j+1)^2 = \sigma_c^2 \sum_{m=1}^T m^2 = \sigma_c^2 \frac{T(T+1)(2T+1)}{6},$$

and dividing by T^2 gives (8). The leading behaviour is $\sigma_c^2(2T^2)/(6T) = T\sigma_c^2/3$. $\square$

Remark. A standard error computed from s_y^2/T is therefore wrong by a factor that grows with the length of the series, and it is wrong in the dangerous direction: too small. Every t statistic built on it is inflated, which is what produces the spurious regressions of §5.2.

Theorem 9.17 (Forecasting a random walk). Let $\hat{\mu}_c = \bar{c} = (y_T - y_0)/T$ be the average change over the observed series. The l -step forecast and its error variance are

$$\hat{y}_{T+l} = y_T + l\bar{c}, \quad \text{var}(y_{T+l} - \hat{y}_{T+l}) = l\sigma_c^2\left(1 + \frac{l}{T}\right). \quad (9)$$

Ignoring the estimation term, an approximate 95% interval is $y_T + l\bar{c} \pm 2s_c\sqrt{l}$, whose width grows like $\sqrt{l}$.

Proof. Iterating (6) forward, $y_{T+l} = y_T + \sum_{k=1}^l c_{T+k}$. Since y_T is known at time T and the future changes have mean μ_c , the mean squared error forecast is $y_T + l\mu_c$, estimated by replacing

μ_c with $\bar{c}$. The error is

$$y_{T+l} - \hat{y}_{T+l} = \sum_{k=1}^l (c_{T+k} - \mu_c) - l(\bar{c} - \mu_c),$$

a difference of two independent terms, the first involving only future changes and the second only past ones. Their variances are $l\sigma_c^2$ and $l^2\sigma_c^2/T$, and adding gives (9). $\square$

Proposition 9.18 (Random walk against linear trend). Writing $c_t = \mu_c + \varepsilon_t$ with $\{\varepsilon_t\}$ white noise, the random walk is

$$y_t = y_0 + \mu_c t + u_t, \quad u_t = \sum_{j=1}^t \varepsilon_j, \quad (10)$$

which has the same deterministic part as the linear trend model (2). The two differ only in the error process: stationary white noise there, a random walk here.

Proof. Substitute $c_j = \mu_c + \varepsilon_j$ into (6) and collect the constant terms. $\square$

Remark. The distinction is not cosmetic. Under (2) a shock dies immediately and the series returns to its trend line; under (10) a shock is absorbed into u_t permanently and the level never returns. That is why the two forecast intervals of §4.4 behave so differently, and why the choice between them cannot be made from R^2 : both fit a rising series well.

2.4 Filtering

Definition 9.19 (Difference operator). $\Delta y_t = y_t - y_{t-1}$, and Δ^d is Δ applied d times. A **filter** is any operation intended to reduce a series to white noise.

Theorem 9.20 (Differencing). (i) If $\{y_t\}$ is a random walk with changes $\{c_t\}$, then $\Delta y_t = c_t$ is white noise.

(ii) If $\{y_t\}$ is itself white noise with variance σ^2 , then Δy_t has variance $2\sigma^2$ and autocorrelations $\rho(1) = -\frac{1}{2}$, $\rho(k) = 0$ for $k \geq 2$.

(iii) If $y_t = \beta_0 + \beta_1 t + \varepsilon_t$ is a linear trend with white noise errors, then $\Delta y_t = \beta_1 + \varepsilon_t - \varepsilon_{t-1}$ is

stationary with the autocorrelations of (ii).

Proof. (i) is the definition (6) rearranged.

(ii) $\text{var}(y_t - y_{t-1}) = \sigma^2 + \sigma^2 = 2\sigma^2$ by independence, and $\text{cov}(\Delta y_t, \Delta y_{t-1}) = \text{cov}(y_t - y_{t-1}, y_{t-1} - y_{t-2}) = -\text{var}(y_{t-1}) = -\sigma^2$, all other cross-covariances vanishing; dividing by $2\sigma^2$ gives $-\frac{1}{2}$. For $k \geq 2$ the two differences involve disjoint sets of observations, so the covariance is 0.

(iii) Differencing removes the linear term and leaves β_1 plus the difference of the errors, which is (ii). $\square$

Remark. Part (ii) is the cost of differencing a series that did not need it: the variance doubles and a negative first-order autocorrelation is introduced where there was none. A first difference is a strong operation, and the diagnostic for whether it was needed is whether it reduced the variability: for a random walk it does, dramatically, and for stationary data it does the opposite.

Proposition 9.21 (Log differences are proportional changes). Let $\pi_t = y_t/y_{t-1} - 1$ be the proportional change. Then $\Delta \ln y_t = \ln(1 + \pi_t)$, and $|\Delta \ln y_t - \pi_t| \leq 2\pi_t^2$ whenever $|\pi_t| \leq \frac{1}{2}$.

Proof. $\ln y_t - \ln y_{t-1} = \ln(y_t/y_{t-1}) = \ln(1 + \pi_t)$. For the bound, Taylor's theorem with Lagrange remainder (prerequisites) applied to $x \mapsto \ln(1 + x)$ at 0 gives $\ln(1 + \pi) = \pi - \pi^2/\{2(1 + \xi)^2\}$ for some ξ between 0 and π . If $|\pi| \leq \frac{1}{2}$ then $\xi \in (-\frac{1}{2}, \frac{1}{2})$, so $(1 + \xi)^2 > \frac{1}{4}$ and the remainder is smaller than $2\pi^2$ in modulus. $\square$

Remark (Identifying a random walk). Three checks, all of which the computation of §4.3 performs. The sample standard deviation should fall sharply on differencing, since the level has variance growing like $t\sigma_\epsilon^2$ while the differences have variance σ_ϵ^2 . The sample autocorrelation function of the level should start near 1 and decay slowly (Theorem 9.15), and that of the differences should be negligible at every lag. And a time series plot of the differences should show no trend and constant spread.

3 Forecast Evaluation and Autocorrelation

FREES §7.6, §8.1.

3.1 Out-of-sample validation

Definition 9.22 (The protocol). Split the series at T_1 , keeping $t = 1, \dots, T_1$ as the **development** subsample and $t = T_1 + 1, \dots, T$ as the **validation** subsample, $T_2 = T - T_1$. Fit each candidate model on the development subsample only. For each validation date t , produce the one-step forecast $\hat{y}_t$ using the fitted model and the observations up to $t - 1$, and record the **forecast residual** $e_t = y_t - \hat{y}_t$.

Remark. The split must respect time. A random split, of the kind Unit 1 used for cross-sectional data, would let a model be fitted on observations later than the ones it predicts, and for a series with a trend that alone can make a useless model look excellent. Nothing dated after $t - 1$ may enter $\hat{y}_t$, the estimated coefficients included.

Definition 9.23 (Comparison statistics). With $e_t = y_t - \hat{y}_t$ over the T_2 validation dates,

$$\text{ME} = \frac{1}{T_2} \sum e_t, \quad \text{MPE} = \frac{100}{T_2} \sum \frac{e_t}{y_t}, \quad \text{MSE} = \frac{1}{T_2} \sum e_t^2, \quad (11)$$

$$\text{MAE} = \frac{1}{T_2} \sum |e_t|, \quad \text{MAPE} = \frac{100}{T_2} \sum \left| \frac{e_t}{y_t} \right|. \quad (12)$$

Proposition 9.24 (What each statistic sees). $\text{MSE} = \text{ME}^2 + \widehat{\text{var}}(e)$, where $\widehat{\text{var}}(e) = T_2^{-1} \sum (e_t - \text{ME})^2$. Hence ME measures only systematic over- or under-prediction and can be near zero for a forecast that is wildly wrong in both directions, while MSE and MAE cannot. MPE and MAPE are the same two statistics on a relative scale and are undefined when some $y_t = 0$.

Proof. $T_2^{-1} \sum e_t^2 = T_2^{-1} \sum \{(e_t - \bar{e}) + \bar{e}\}^2 = T_2^{-1} \sum (e_t - \bar{e})^2 + \bar{e}^2$, the cross term vanishing because $\sum (e_t - \bar{e}) = 0$; and $\bar{e} = \text{ME}$. $\square$

3.2 Autocorrelation

Definition 9.25 (Sample autocorrelation). For $k \geq 1$,

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}. \quad (13)$$

The plot of r_k against k is the **correlogram**.

Proposition 9.26 (Range). $|r_k| \leq 1$.

Proof. Extend the sum in the numerator to all t by setting $y_t - \bar{y} = 0$ for $t \leq 0$, and apply Cauchy–Schwarz (prerequisites) to the vectors with entries $y_t - \bar{y}$ and $y_{t-k} - \bar{y}$: the numerator is at most $\{\sum_t (y_t - \bar{y})^2\}^{1/2} \{\sum_t (y_{t-k} - \bar{y})^2\}^{1/2}$, and each factor is at most $\{\sum_t (y_t - \bar{y})^2\}^{1/2}$. $\square$

Theorem 9.27 (The correlogram of white noise). Let $y_1, \dots, y_T$ be white noise with mean μ

and variance σ^2 , and fix $k \geq 1$. Put

$$\tilde{r}_k = \frac{1}{(T-k)\sigma^2} \sum_{t=k+1}^T (y_t - \mu)(y_{t-k} - \mu),$$

the version with the mean known. Then the summands are uncorrelated with mean zero, and

$$\mathbf{E}[\tilde{r}_k] = 0, \quad \text{var}(\tilde{r}_k) = \frac{1}{T-k}. \quad (14)$$

Proof. Write $\xi_t = y_t - \mu$ and $z_t = \xi_t \xi_{t-k}$. Independence gives $\mathbf{E}[z_t] = \mathbf{E}[\xi_t] \mathbf{E}[\xi_{t-k}] = 0$ and $\mathbf{E}[z_t^2] = \mathbf{E}[\xi_t^2] \mathbf{E}[\xi_{t-k}^2] = \sigma^4$.

For $t \neq u$ the product $z_t z_u = \xi_t \xi_{t-k} \xi_u \xi_{u-k}$ involves at least one index appearing exactly once: the four indices $t, t-k, u, u-k$ can coincide only through $t = u-k$ or $u = t-k$, and in either case exactly two of them coincide and the remaining two are distinct from each other and from the pair. Factorising the expectation over independent coordinates therefore leaves a factor $\mathbf{E}[\xi] = 0$, so $\mathbf{E}[z_t z_u] = 0$ and the z_t are uncorrelated.

Hence $\text{var}(\sum_{t>k} z_t) = (T - k)\sigma^4$, and dividing by $(T - k)\sigma^2$ gives (14). □

Corollary 9.28. Under white noise, $\mathbf{P}\{|\tilde{r}_k| > 2/\sqrt{T - k}\} \leq \frac{1}{4}$. The horizontal lines at $\pm 2/\sqrt{T}$ drawn on a correlogram are two standard deviations of $\tilde{r}_k$.

Proof. Chebyshev's inequality (prerequisites) with (14). □

Remark. Chebyshev gives only $\frac{1}{4}$; the conventional reading of the band as a 95% device rests on a normal approximation for $\tilde{r}_k$, which is not proved here. The sequence $z_t = \xi_t \xi_{t-k}$ is not independent, so the central limit theorem of the prerequisites does not apply to it directly. What §4.5 does instead is measure the coverage by simulation, and it comes out at the advertised 95%.

Remark. Replacing μ by $\bar{y}$ and $(T - k)$ by T , as in (13), introduces a small negative bias: $\mathbf{E}[r_k] \approx -1/T$, which §4.5 also exhibits. For a series of any length the bias is negligible beside the standard deviation $1/\sqrt{T}$.

3.3 Autocorrelated errors in a regression

Definition 9.29 (Durbin–Watson). For residuals $e_1, \dots, e_T$ from a fitted regression,

$$\text{DW} = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=1}^T e_t^2}. \quad (15)$$

Proposition 9.30 (Exact relation to the first autocorrelation). With $r_1 = \sum_{t=2}^T e_t e_{t-1} / \sum_{t=1}^T e_t^2$,

$$\text{DW} = 2(1 - r_1) - \frac{e_1^2 + e_T^2}{\sum_{t=1}^T e_t^2}. \quad (16)$$

So DW is near 2 when the residuals are uncorrelated, near 0 under strong positive autocorrelation, and near 4 under strong negative autocorrelation.

Proof. Expanding the numerator of (15) and writing $S = \sum_{t=1}^T e_t^2$,

$$\sum_{t=2}^T (e_t - e_{t-1})^2 = \sum_{t=2}^T e_t^2 + \sum_{t=2}^T e_{t-1}^2 - 2 \sum_{t=2}^T e_t e_{t-1} = (S - e_1^2) + (S - e_T^2) - 2r_1 S.$$

Dividing by S gives (16). The three cases follow by putting $r_1 = 0, 1, -1$ and noting the correction term is $O(1/T)$ when no single residual dominates. $\square$

Remark. Positive residual autocorrelation is the normal state of affairs for a regression on time series data, and its consequence is the one Theorem 9.16 exhibits in extreme form: the reported standard errors are too small, so significance is overstated. The remedies are to model the dependence, which is Unit 10, or to difference the series, which §2.4 does.

4 Computation: Formulas Against Package Output

FREES §7.2, §7.4, §7.6.

Two series from base R. **AirPassengers**: monthly international airline passengers, $T = 144$, with a rising trend and a seasonal swing that grows with the level. **EuStockMarkets**: daily closing values

of four European indices; the DAX column is used, $T = 1860$.

```
1 suppressMessages(library(lmtest))
2 y <- as.numeric(AirPassengers); T <- length(y); tt <- seq_len(T)
3 mo <- factor(rep(month.abb, length.out = T), levels = month.abb)
4 dax <- as.numeric(EuStockMarkets[, "DAX"])
5 c(T_air = T, T_dax = length(dax), mean_air = mean(y), sd_air = sd(y))
```

```
##      T_air      T_dax mean_air      sd_air
##      144.0     1860.0     280.3     120.0
```

4.1 Trend and season (§1.2–§1.3)

```
1 m1 <- lm(y ~ tt)                # linear trend only
2 m2 <- lm(y ~ tt + mo)           # additive trend and season
3 m3 <- lm(log(y) ~ tt + mo)      # multiplicative, on the log scale
4 c(R2_trend = summary(m1)$r.squared, R2_trend_season = summary(m2)$r.squared,
5   R2_log = summary(m3)$r.squared,
6   s_trend = summary(m1)$sigma, s_trend_season = summary(m2)$sigma)
```

```
##      R2_trend R2_trend_season      R2_log
##      0.8536      0.9559      0.9835
```

```
##          s_trend  s_trend_season
##          46.0572      26.3303
```

Theorem 9.4 gives the sampling variances in closed form.

```
1 X <- cbind(1, tt); V <- solve(crossprod(X))
2 c(var_b1_matrix = V[2, 2], formula = 12/(T*(T^2 - 1)),
3   var_b0_matrix = V[1, 1], formula_b0 = 2*(2*T + 1)/(T*(T - 1)),
4   cov_matrix = V[1, 2], formula_cov = -6/(T*(T - 1)))
```

```
## var_b1_matrix      formula var_b0_matrix      formula_b0
## 0.000004019  0.000004019  0.028069153  0.028069153
## cov_matrix      formula_cov
## -0.000291375 -0.000291375
```

Propositions 9.6 and 9.7: the dummies alone fit the seasonal means, and the harmonic basis of Theorem 9.8 spans the same space.

```
1 ms <- lm(y ~ mo - 1)
2 ph <- 2*pi*tt/12
3 Fh <- cbind(1, sapply(1:5, function(j) cos(j*ph)),
4             sapply(1:5, function(j) sin(j*ph)), cos(6*ph))
5 D  <- model.matrix(~ mo - 1)
```

```
6 c(dummies_are_means = max(abs(coef(ms) - tapply(y, mo, mean))),
7   rank_dummies = qr(D)$rank, rank_harmonics = qr(Fh)$rank,
8   rank_of_both_together = qr(cbind(D, Fh))$rank,
9   fitted_values_differ_by = max(abs(fitted(lm(y ~ tt + mo)) -
10                                     fitted(lm(y ~ tt + Fh - 1)))))
```

```
##          dummies_are_means          rank_dummies
##          1.137e-13          1.200e+01
##          rank_harmonics    rank_of_both_together
##          1.200e+01          1.200e+01
## fitted_values_differ_by
##          1.094e-12
```

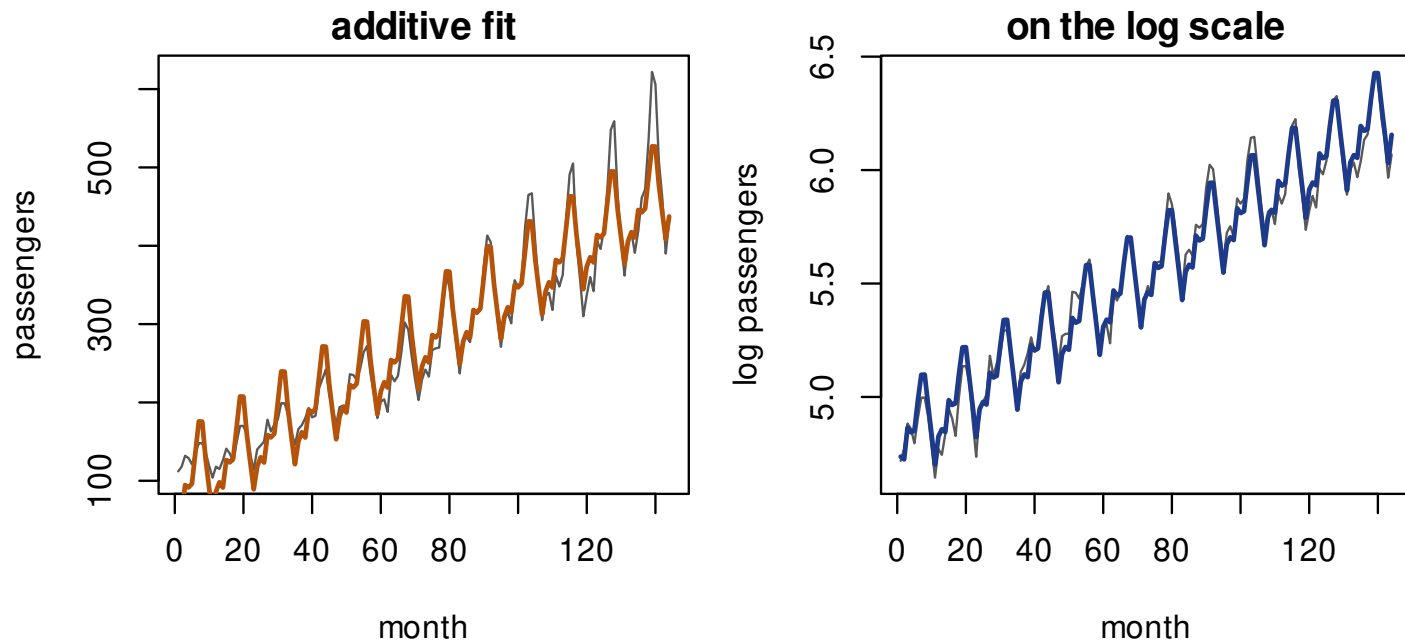


Figure 9.1: A linear trend plus twelve monthly dummies. On the original scale the fitted seasonal amplitude is constant while the observed one grows, so the fit is too wide early and too narrow late; on the log scale the amplitude is constant by construction (Proposition 9.2) and the same model fits throughout.

4.2 Extrapolating a polynomial

```

1 t(sapply(c(1, 2, 4, 8), function(dg) {
2   m <- lm(y ~ poly(tt, dg))
3   c(degree = dg, R2 = summary(m)$r.squared,
4     forecast_t_180 = predict(m, data.frame(tt = 180))) })))

```

```

##      degree      R2 forecast_t_180.1
## [1,]      1 0.8536             565.9
## [2,]      2 0.8618             634.8
## [3,]      4 0.8621             630.2
## [4,]      8 0.8699          -19372.6

```

Each increase in degree raises R^2 and the degree-8 fit, the best of the four by that measure, forecasts a negative number of passengers three years ahead.

4.3 Is the DAX a random walk? (§2.4)

```

1 d1 <- diff(dax); d1 <- diff(log(dax))
2 c(sd_level = sd(dax), sd_difference = sd(d1),
3   ratio = sd(dax)/sd(d1),
4   acf1_level = acf(dax, plot = FALSE)$acf[2],

```

```
5 acf1_difference = acf(d1, plot = FALSE)$acf[2],  
6 acf1_log_difference = acf(d1, plot = FALSE)$acf[2])
```

##	sd_level	sd_difference	ratio
##	1084.7927403	32.4972571	33.3810555
##	acf1_level	acf1_difference	acf1_log_difference
##	0.9973841	0.0008130	-0.0004346

All three checks of §2.4 point the same way: differencing cuts the standard deviation by a factor of 33, the level has first autocorrelation 0.997, and the differences have 0.0008.

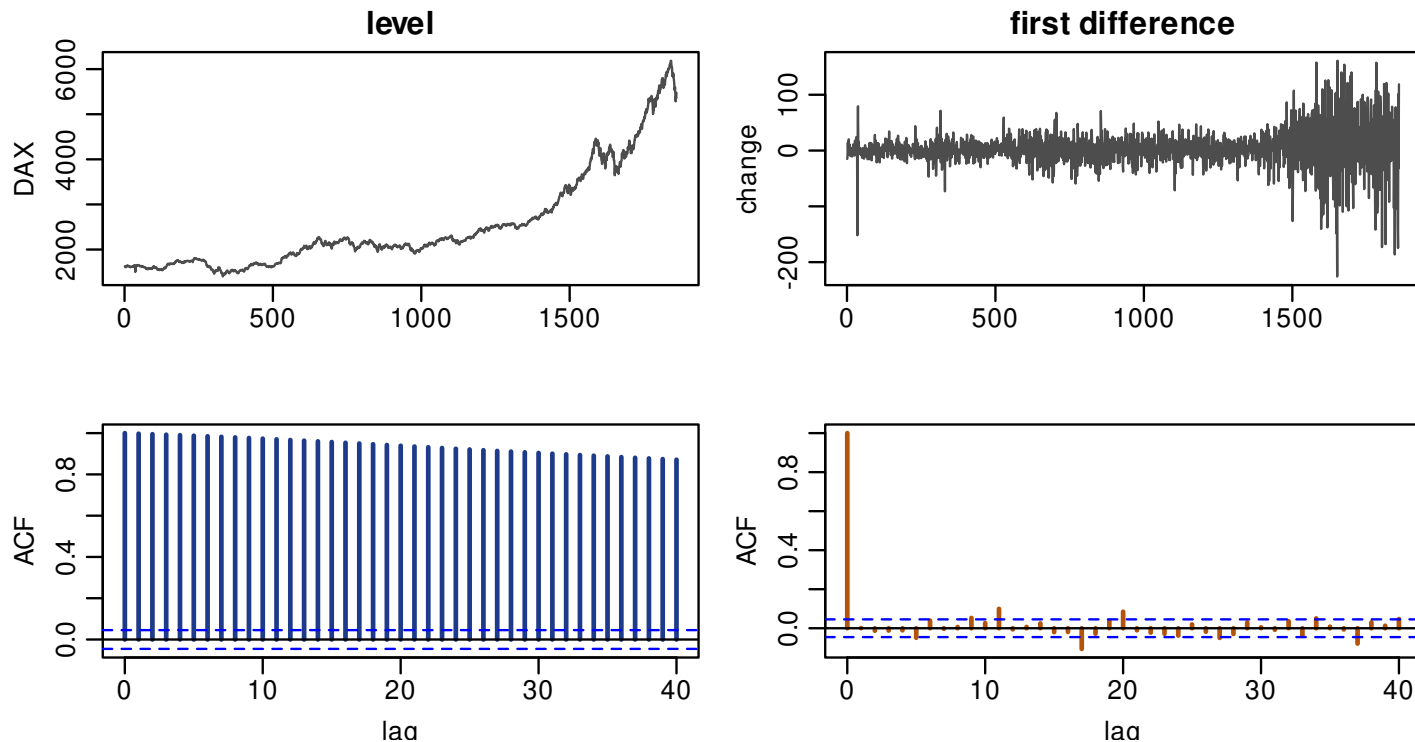


Figure 9.2: The signature of a random walk. The level wanders with growing spread and its correlogram decays hardly at all over forty lags, as $\text{corr}(y_{t-k}, y_t) = \sqrt{1 - k/t}$ of Theorem 9.15 predicts; the differences look like white noise.

4.4 The two forecast intervals (§1.4, §2.3)

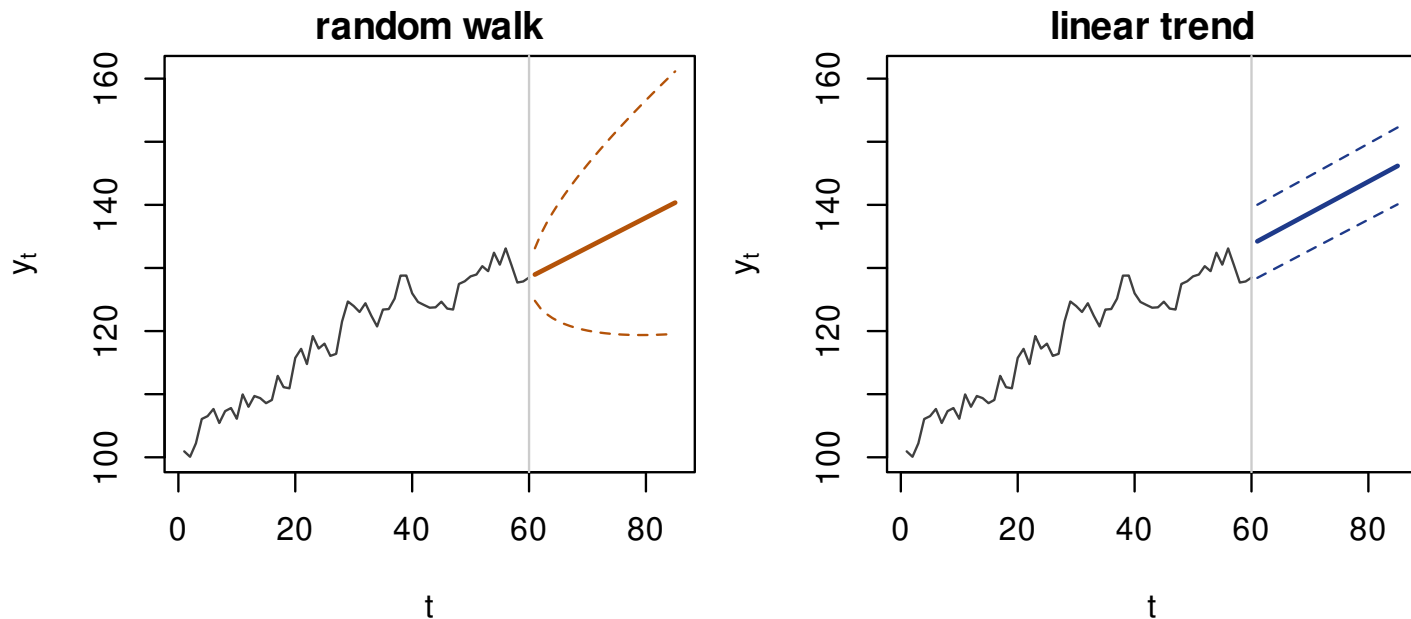


Figure 9.3: One simulated series, two models. The random walk forecasts from the last observation with an interval widening like $\sqrt{t}$ (Theorem 9.17); the linear trend forecasts from the fitted line with an interval of almost constant width (Proposition 9.9). Both fit the observed part well.

Theorem 9.16 says the variance of the sample mean of a random walk grows with T , while the estimate computed as though the data were independent does not.

```

1 set.seed(4); B <- 3000
2 t(sapply(c(50, 200, 800), function(Tn) {
3   r <- replicate(B, { z <- cumsum(rnorm(Tn)); c(mean(z), var(z)/Tn) })
4   c(T = Tn, simulated_var_of_ybar = var(r[1, ]),
5     formula = (Tn + 1)*(2*Tn + 1)/(6*Tn),
6     mean_of_naive_estimate = mean(r[2, ])) })))

```

```

##           T simulated_var_of_ybar formula mean_of_naive_estimate
## [1,]   50              17.16    17.17              0.1722
## [2,]  200              67.42    67.17              0.1705
## [3,]  800             277.88   267.17              0.1682

```

```

1 set.seed(7); w <- rnorm(20000)
2 c(var_white_noise = var(w), var_of_its_difference = var(diff(w)),
3   ratio = var(diff(w))/var(w),
4   acf1_of_difference = acf(diff(w), plot = FALSE)$acf[2], theory = -0.5)

```

```

##           var_white_noise var_of_its_difference
##              1.0116              1.9971

```

```
##          ratio      acf1_of_difference
##      1.9741      -0.4965
##      theory
##     -0.5000
```

4.5 Correlograms and Durbin–Watson (§3.2–§3.3)

Theorem 9.27 gives the mean and variance of r_k under white noise; the 95% calibration of the $\pm 2/\sqrt{T}$ band is not proved there and is measured here.

```
1 set.seed(3); B <- 5000
2 t(sapply(c(50, 200), function(Tn) {
3   rk <- replicate(B, acf(rnorm(Tn), plot = FALSE, lag.max = 1)$acf[2])
4   c(T = Tn, mean_r1 = mean(rk), minus_one_over_T = -1/Tn,
5     sd_r1 = sd(rk), one_over_sqrt_T = 1/sqrt(Tn),
6     inside_the_band = mean(abs(rk) < 2/sqrt(Tn))) })))
```

```
##          T  mean_r1 minus_one_over_T  sd_r1 one_over_sqrt_T
## [1,]  50 -0.018786      -0.020 0.14015      0.14142
## [2,] 200 -0.004112      -0.005 0.07037      0.07071
##      inside_the_band
## [1,]          0.9556
```

```
## [2,]          0.9576
```

```
1 set.seed(5); n <- 60
2 xg <- rnorm(n); eg <- as.numeric(arima.sim(list(ar = 0.6), n))
3 md <- lm(I(1 + 2*xg + eg) ~ xg); rr <- residuals(md)
4 r1 <- sum(rr[-1]*rr[-n])/sum(rr^2)
5 c(DW_manual = sum(diff(rr)^2)/sum(rr^2),
6   DW_package = unname(dwtest(md)$statistic),
7   approximation_2_1_minus_r1 = 2*(1 - r1),
8   exact_identity = unname(2*(1 - r1) - (rr[1]^2 + rr[n]^2)/sum(rr^2)))
```

```
##              DW_manual              DW_package
##              1.274              1.274
## approximation_2_1_minus_r1      exact_identity
##              1.360              1.274
```

The approximation $2(1 - r_1)$ is off by 0.09 on sixty observations; the exact identity (16) matches the package to the last digit.

4.6 Comparing forecasts out of sample (§3.1)

Three candidates for **AirPassengers**, fitted on the first 120 months and judged on the last 24: a trend with monthly dummies, a random walk on the level, and a random walk on the same month a year earlier.

```

1 T1 <- 120; ya <- y[(T1 + 1):T]
2 fit_ts <- lm(y ~ tt + mo, subset = 1:T1)
3 f_ts <- predict(fit_ts, newdata = data.frame(tt = (T1 + 1):T,
4                                             mo = mo[(T1 + 1):T]))
5 f_rw <- y[T1:(T - 1)] + mean(diff(y[1:T1]))
6 f_srw <- y[(T1 + 1 - 12):(T - 12)] + mean(diff(y[1:T1], lag = 12))
7 stats <- function(e, a) c(ME = mean(e), MPE = 100*mean(e/a), MSE = mean(e^2),
8                           MAE = mean(abs(e)), MAPE = 100*mean(abs(e/a)))
9 round(rbind(trend_plus_season = stats(ya - f_ts, ya),
10          random_walk = stats(ya - f_rw, ya),
11          seasonal_random_walk = stats(ya - f_srw, ya)), 4)

```

##	ME	MPE	MSE	MAE	MAPE
## trend_plus_season	26.139	4.5904	2298.6	34.64	6.880
## random_walk	2.068	-0.0067	2670.0	43.89	9.675

```
## seasonal_random_walk 19.324  4.1143  607.9 21.77 4.743
```

The plain random walk has the smallest ME and the largest MSE: it is unbiased on average and wrong every month, because it carries no seasonal term. Proposition 9.24 is the arithmetic behind that.

```
1 e <- ya - f_rw
2 c(MSE = mean(e^2), ME_squared = mean(e)^2,
3   variance_term = mean((e - mean(e))^2),
4   difference = mean(e^2) - mean(e)^2 - mean((e - mean(e))^2))
```

```
##           MSE      ME_squared variance_term      difference
##      2669.981         4.275      2665.707         0.000
```

5 Pitfalls

5.1 Choosing the polynomial degree by fit

§4.2 showed a degree-8 trend with the best R^2 of its family forecasting a negative passenger count. The same comparison out of sample reverses the ranking.

```
1 T1 <- 120; ya <- y[(T1 + 1):T]
2 t(sapply(c(1, 2, 4, 8), function(dg) {
```

```

3 m <- lm(y ~ poly(tt, dg), subset = 1:T1)
4 f <- predict(m, newdata = data.frame(tt = (T1 + 1):T))
5 c(degree = dg, R2_in_sample = summary(m)$r.squared,
6   MSE_out_of_sample = mean((ya - f)^2)) })

```

```

##      degree R2_in_sample MSE_out_of_sample
## [1,]      1      0.8356           5593
## [2,]      2      0.8418           4779
## [3,]      4      0.8461          15540
## [4,]      8      0.8535         4943287

```

Diagnostic. A trend chosen on R^2 , and forecasts that leave the range of the data faster than the data ever moved. **Fix.** Judge the degree on out-of-sample error (§3.1). A polynomial is an interpolating device; nothing in it knows that a passenger count is positive.

5.2 Regressing one random walk on another

Theorem 9.16 says the usual standard error is too small under a random walk, by a factor that grows with T . The consequence is that two entirely unrelated series can look strongly related.

```

1 set.seed(2); B <- 2000
2 spur <- function(Tn, difference) t(replicate(B, {

```

```

3   a <- cumsum(rnorm(Tn)); b <- cumsum(rnorm(Tn))
4   if (difference) { a <- diff(a); b <- diff(b) }
5   m <- lm(a ~ b)
6   c(summary(m)$coefficients[2, 3], summary(m)$r.squared) )))
7   tab <- function(difference) t(sapply(c(50, 200, 800), function(Tn) {
8     r <- spur(Tn, difference)
9     c(T = Tn, mean_abs_t = mean(abs(r[, 1])),
10      reject_at_5_percent = mean(abs(r[, 1]) > 1.96),
11      median_R2 = median(r[, 2])) )))
12   tab(FALSE)

```

```

##           T mean_abs_t reject_at_5_percent median_R2
## [1,]  50      4.037           0.6865      0.1817
## [2,] 200      8.038           0.8350      0.1781
## [3,] 800     15.415           0.9105      0.1571

```

```

1   tab(TRUE)

```

```

##           T mean_abs_t reject_at_5_percent median_R2
## [1,]  50      0.8014           0.0545 0.0093543
## [2,] 200      0.8092           0.0480 0.0023854

```

```
## [3,] 800      0.7779      0.0435 0.0005003
```

Diagnostic. A regression between two trending series with a large t statistic, a respectable R^2 , and residuals whose Durbin–Watson statistic is far below 2. **Fix.** Difference both series first, or model the dependence directly (Unit 10). The rejection rate returns to its nominal 5% and the mean $|t|$ to the $\sqrt{2/\pi} = 0.798$ of a standard normal.

5.3 Splitting a series at random

```
1 set.seed(8)
2 oos <- function(random) {
3   idx <- if (random) sort(sample(T, T1)) else 1:T1
4   m <- lm(y ~ tt + mo, subset = idx)
5   te <- setdiff(seq_len(T), idx)
6   mean((y[te] - predict(m, data.frame(tt = tt[te], mo = mo[te]))))^2)
7 }
8 c(chronological_split = oos(FALSE), random_split = oos(TRUE),
9   ratio = oos(FALSE)/oos(TRUE))
```

```
## chronological_split      random_split      ratio
##           2298.628           1128.662           1.636
```

Diagnostic. A validation set containing dates earlier than some of the training dates. Here the random split reports 1,129 against the 2,299 a genuine forecast achieves, so the model looks 1.6 times better than it is. **Fix.** Split on time. A random split lets the fit interpolate between neighbouring months instead of extrapolating past the end of the series, which is the only thing a forecast ever has to do.

5.4 Differencing a series that did not need it

```

1 set.seed(12); z <- 5 + rnorm(400)           # stationary already
2 t(sapply(0:2, function(d) {
3   u <- if (d == 0) z else diff(z, differences = d)
4   c(differences = d, variance = var(u),
5     acf1 = acf(u, plot = FALSE)$acf[2]) })))

```

##	differences	variance	acf1
## [1,]	0	0.9105	-0.04765
## [2,]	1	1.9070	-0.51308
## [3,]	2	5.7611	-0.65470

Diagnostic. A first autocorrelation near $-\frac{1}{2}$ after differencing, and a variance larger than before. **Fix.** Difference only when the level wanders. Theorem 9.20(ii) says that differencing white noise

doubles its variance and manufactures a negative autocorrelation; a second difference triples it again. The test is whether differencing reduces the standard deviation, not whether the result looks flat.

6 Exercises

Exercise 9.1. A monthly series has a seasonal swing of about 10% of its level, and the level has doubled over the sample. Should the seasonal component be modelled additively or multiplicatively, and what does a logarithmic transform do?

Solution. Multiplicatively: an additive model would fit one constant swing, too wide at the start of the sample and too narrow at the end, exactly the pattern in the left panel of Figure 9.1. By Proposition 9.2, taking logarithms turns the multiplicative decomposition into an additive one and turns the constant proportional swing into a constant additive one, $\ln(1.10) = 0.0953$ on the log scale whatever the level. A linear trend plus seasonal dummies fitted to $\ln y_t$ therefore has a constant seasonal amplitude by construction, and back-transforming gives a swing proportional to the level. ■

Exercise 9.2. Derive $\text{var}(\hat{\beta}_1)$ for the linear trend in time model and compare its rate in T with that of a slope on an ordinary covariate.

Solution. Theorem 9.4: $\det(\mathbf{X}'\mathbf{X}) = T^2(T^2 - 1)/12$ and the lower-right entry of the inverse is $T/\det$, giving $\text{var}(\hat{\beta}_1) = 12\sigma^2/\{T(T^2 - 1)\}$, of order $\sigma^2 T^{-3}$.

For an ordinary covariate, $\text{var}(\hat{\beta}_1) = \sigma^2/\sum_i (x_i - \bar{x})^2$, and if the x_i are drawn from a fixed distribution then $\sum_i (x_i - \bar{x})^2$ grows like T , giving order $\sigma^2 T^{-1}$. The extra two powers come from the design spreading out: the range of t grows with T instead of staying fixed. The gain is genuine only when the errors really are uncorrelated. ■

Exercise 9.3. A quarterly series is modelled with an intercept and four quarterly dummies. What goes wrong, and what are the two standard repairs? What does the harmonic alternative buy?

Solution. The four dummies sum to the intercept column, so the design has rank 4 with 5 columns and the coefficients are not identified (Proposition 9.6). Drop the intercept, in which case each $\hat{\alpha}_j$ is the quarterly mean (Proposition 9.7), or drop one dummy, in which case the remaining coefficients are contrasts with the omitted quarter. The fitted values are identical.

By Theorem 9.8 the harmonics $\{1, \cos(\pi t/2), \sin(\pi t/2), \cos(\pi t)\}$ span the same four-dimensional space, so they fit identically too. What they buy is the option of dropping the highest frequency, which imposes a smooth seasonal shape at the cost of one parameter; with $s = 4$ that saving is small, with $s = 12$ or $s = 52$ it is not. ■

Exercise 9.4. Show that a random walk is not stationary, and compute $\text{corr}(y_{t-1}, y_t)$ at $t = 100$. What does this imply about its correlogram?

Solution. By Theorem 9.15, $\text{var}(y_t) = t\sigma_c^2$ depends on t , so the second requirement of Definition 9.10 fails; if $\mu_c \neq 0$ the mean depends on t as well.

$\text{corr}(y_{99}, y_{100}) = \sqrt{99/100} = 0.995$. More generally $\text{corr}(y_{t-k}, y_t) = \sqrt{1 - k/t}$, which at $t = 100$ is 0.95 even at lag $k = 10$. The sample correlogram therefore starts near 1 and decays almost imperceptibly over the first several lags, which is the pattern in the lower-left panel of Figure 9.2 and the usual visual test for a random walk. ■

Exercise 9.5. A series identified as a random walk ends at $y_T = 250$ with average change $\bar{c} = 1.4$ and $s_c = 6$ over $T = 80$ observations. Forecast y_{T+4} and y_{T+16} with approximate 95% intervals.

By what factor does the interval widen?

Solution. By Theorem 9.17 the point forecasts are $250 + 4(1.4) = 255.6$ and $250 + 16(1.4) = 272.4$. The half-widths are $2s_c\sqrt{l} = 2(6)\sqrt{4} = 24.0$ and $2(6)\sqrt{16} = 48.0$, giving $(231.6, 279.6)$ and $(224.4, 320.4)$.

The interval widens by $\sqrt{16/4} = 2$. The exact error variance $l\sigma_c^2(1 + l/T)$ inflates these slightly, by $\sqrt{1 + 4/80} = 1.025$ and $\sqrt{1 + 16/80} = 1.095$, the estimation of the drift mattering more the further ahead the forecast goes. ■

Exercise 9.6. Show that the variance of the sample mean of a driftless random walk grows with T , and say what it implies for regressions on trending series.

Solution. Theorem 9.16: c_j enters $T - j + 1$ of the y_t , so $\text{var}(\sum_t y_t) = \sigma_c^2 \sum_m m^2 = \sigma_c^2 T(T + 1)(2T + 1)/6$ and $\text{var}(\bar{y}) = \sigma_c^2 (T + 1)(2T + 1)/(6T) \sim T\sigma_c^2/3$.

The usual formula s_y^2/T presumes independence and does not grow. Every standard error built on that presumption is therefore too small, by a factor increasing with the length of the series, so t statistics are inflated without limit. §5.2 shows two independent random walks rejecting the null

of no relation 91% of the time at $T = 800$. ■

Exercise 9.7. What does differencing do to a random walk, to white noise, and to a linear trend with white noise errors? How would you tell whether a difference was needed?

Solution. Theorem 9.20. A random walk becomes its own white noise increments. White noise becomes a series of doubled variance with first autocorrelation $-\frac{1}{2}$ and nothing beyond. A linear trend becomes the constant β_1 plus the same differenced-noise structure, so the trend is removed but the same penalty is paid.

The test is the standard deviation: for a random walk it falls sharply, by a factor of 33 on the DAX in §4.3, and for an already stationary series it rises by $\sqrt{2}$. A first autocorrelation close to $-\frac{1}{2}$ in the differenced series is the signature of having differenced once too often. ■

Exercise 9.8. Two forecasting methods are compared over 24 validation months. Method A has $ME = 0.1$ and $MSE = 900$; method B has $ME = 8$ and $MSE = 100$. Which is better, and what does the comparison show about ME?

Solution. Method B, decisively: its typical error is about $\sqrt{100} = 10$ against $\sqrt{900} = 30$. By Proposition 9.24, $\text{MSE} = \text{ME}^2 + \widehat{\text{var}}(e)$, so A has $\widehat{\text{var}}(e) = 900 - 0.01 = 899.99$ and B has $100 - 64 = 36$. A is unbiased and wildly variable; B is biased by 8 and otherwise accurate. ME measures only the systematic part and is near zero whenever errors of both signs cancel, so it must never be used alone. The right reading of B's result is that a bias of 8 has been detected and can be removed, after which its MSE would be about 36. ■

Exercise 9.9. For white noise of length $T = 200$, what are the mean and standard deviation of r_5 ? What do the lines at $\pm 2/\sqrt{T}$ on a correlogram mean, and what do they not mean?

Solution. By Theorem 9.27 the known-mean version has mean 0 and variance $1/(T - k) = 1/195$, so a standard deviation of 0.0716; with the mean estimated, the mean of r_5 is about $-1/T = -0.005$ and the standard deviation about $1/\sqrt{200} = 0.0707$. §4.5 measures -0.004 and 0.070 .

The lines are two standard deviations of r_k under the hypothesis that the series is white noise. What they are **not** is a proved 95% band: Chebyshev gives only $\mathbf{P}\{|r_k| > 2/\sqrt{T}\} \leq \frac{1}{4}$, and the

usual 95% reading rests on a normal approximation that the prerequisites do not establish for this statistic. Simulation puts the coverage at 95.6% and 95.8% for $T = 50$ and $T = 200$. A second caution: with 20 lags plotted, about one point outside the band is expected even under white noise. ■

Exercise 9.10. A regression on quarterly data over 15 years gives $DW = 0.8$. Estimate the first residual autocorrelation and say what it implies for the reported standard errors.

Solution. By Proposition 9.30, $DW \approx 2(1 - r_1)$ up to a term of order $1/T$, so $r_1 \approx 1 - DW/2 = 1 - 0.4 = 0.6$: strong positive residual autocorrelation over $T = 60$ quarters.

The reported standard errors assume uncorrelated errors and are therefore too small, and the t statistics too large, for the reason of Theorem 9.16: positively correlated errors carry less information than the same number of independent ones, so the effective sample size is smaller than 60. The fit should be redone on differences, or the dependence modelled directly as in Unit 10. ■

Exercise 9.11. In R: (a) verify the closed-form variance of a linear trend slope; (b) show that seasonal dummies and harmonics give identical fits; (c) show that the t statistic from regressing

one random walk on another diverges, and that differencing repairs it.

Solution. (a) is §4.1, (b) §4.1 and (c) §5.2. In (c) the point to watch is that the divergence is in the **standard error**, not the coefficient: the fitted slope between two independent walks is centred at zero, and it is the denominator that shrinks too fast. That is why the rejection rate rises with T instead of settling at 5%, and why differencing, which restores stationarity, restores the nominal size. ■